M5470 : Introduction to Parallel Scientific Computing

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Homework 2

Due Date and Time: 10 February 2025, 11 pm

Question 1: Run the serial version of the code 'pi_ser.c' that implements the Monte-Carlo method for computing the value of π and carry out the following tasks.

(a) Run the code for $n = 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096$ and 8192. Plot the error as a function of n and mention the order of accuracy.

(b) Parallelize the main loop paying attention to variable scoping, scheduling and reduction clauses. In your report, mention which variables that are accessed within that loop will be shared variables and which will be thread-private.

(c) For a large number, say $n = 8192$, run the parallel code with number of threads $p = 2, 4, 8$. For each p, report the error in exponential format up to 16 decimals. Comment on whether the error is identical or different for different p and, if it is different, in what decimal place the differences are seen.

Question 2: Study the code for gradient descent and parallelize it using appropriate OpenMP directives. Briefly mention which loops are amenable to parallelization under what conditions. For all data points, i.e. $m = 10000$, run the code on different number of threads, $p = 2, 4, 8$, and ensure that the loss function value at convergence is identical. Mention in which decimal place you see differences in the loss function value.

Question 3: Study and parallelize the serial code for solving the 2D unsteady heat conduction problem. Run the code for $n = 400$. Plot contours of the temperature at 4 different time instants including the initial and final times. For the final time instant, ensure that the solution obtained with

$p = 2, 4, 8$ differs from the serial solution only at the level of machine precision.

General Instructions:

1. Please prepare a short report with any plots or figures you generated with very brief (2 or 3 lines) comments. Do not waste time and effort in re-typing the question.

2. Submit the pdf on google classroom.

3. Upload your codes as well as the report pdf to Github.